Week 14 Recitation

TA Suha Kim

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Four Market Structures

Exhibit 14.11 Four Market Structures

	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of Firms/ Sellers/Producers	Many	Many	A few	One
Type of Product/ Service Sold	Identical (homogeneous)	Slightly differentiated	Identical or differentiated	Single, undifferentiated product or service
Example of Product	Corn grown by various farmers	Books; CDs	Oil (identical); cars (differentiated)	Patented drugs; tap water
Barriers to Entry	None; free entry and exit	None; free entry and exit	Yes	Yes; high
Price-Taker or Price-Maker?	Price-taker; price given by the market	Price-maker; with a recognition of other sellers	Price-maker; with a strong recognition of other sellers	Price-maker; no competitors, no perfect substitutes

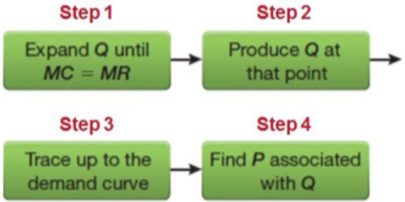
Four Market Structures

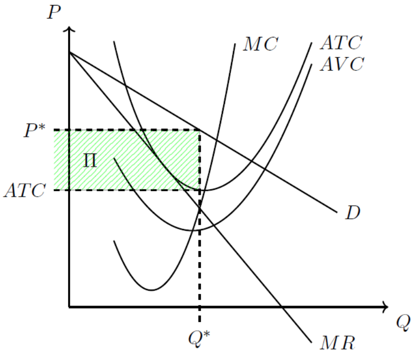
Exhibit 14.11 Four Market Structures

	Perfect Competition	**Monopolistic Competition**	**Oligopoly**	**Monopoly**
Price	$P = MR = MC$	Set $P > MR = MC$	Set $P > MR = MC$, or $P = MR = MC$, depending on type of competition and product differentiation	Set $P > MR = MC$
Residual Demand Curve	Horizontally sloped; perfectly elastic demand curve	Downward-sloping; slightly differentiated products are Available	Downward-sloping	Downward-sloping
Social Surplus	Maximized	Not maximized, but society might benefit from product diversity	Not maximized	Not maximized, but sometimes society benefits from research and development
Long-run Profits	Zero	Zero	Zero or more than zero	More than zero

Review – Monopoly

Monopolist's Problem - From the graph





Monopolistic competition

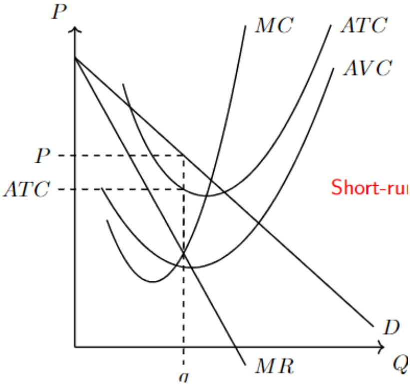
Monopolistic competition is a market structure with:

- Many firms, Free entry and exit
- Firms produce **differentiated products** that are close substitutes to one another: Source of market power
- Because firms face a downward-sloping **residual demand curve** (arises from differentiated products), monopolistic competitive markets are **not efficient**
  - **residual demand curve**: the demand for a firm's good that all other firms do not satisfy, and will vary depending on the prices that all other firm's set

Examples:
Cereal: Kellogg's Rice Krispies (12oz), \$3.49; General Mills Cinnamon Toast Crunch (12oz), \$3.29; Kroger Crispy Rice (12oz), \$1.49; Kellogg's Special K Original (12oz), \$4.49

Monopolistic competition
Short run

Solving the firm's problem in **short-run**:



Firms that have market power(price setting ability) do not have a supply curve

Monopolistic competition

Short run

Solving the firm's problem in **short-run**:

(Summary in words)

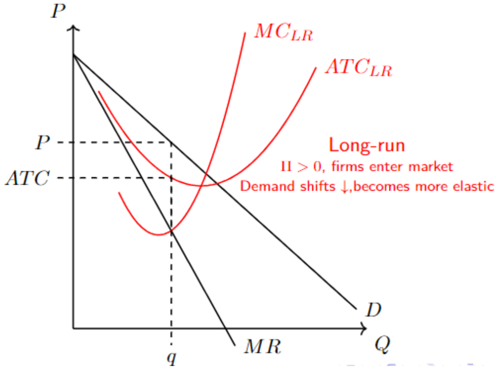
1. Expand production until MR = MC for the last unit produced
2. Set P equal to consumers' willingness to pay for the last unit produced
3. Assess profits at q∗ against profit from shutting down
    - Produce q^* if $\Pi \geq 0$
    - Produce q^* if $-FC \leq \Pi < 0$, i.e., produce as long as $P \geq AVC$
    - Shutdown $(q = 0)$ if $\Pi < -FC$, i.e., $P < AVC$

Firms that have market power(price setting ability) do not have a supply curve

Monopolistic competition

Long run

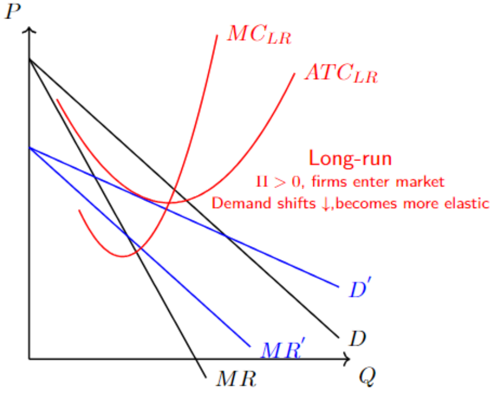
Solving the firm's problem in **Long-run**:



Monopolistic competition

Long run

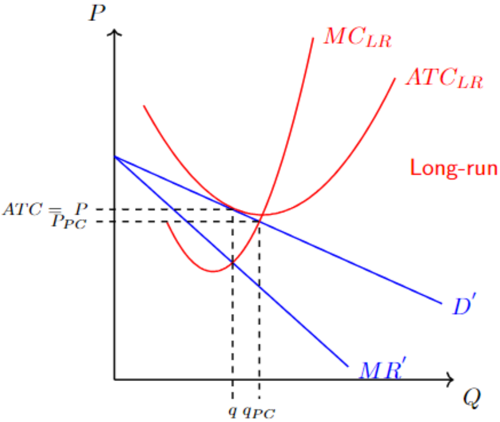
Solving the firm's problem in **Long-run**:



Monopolistic competition

Long run

Solving the firm's problem in **Long-run**:



Monopolistic competition

Long run

Solving the firm's problem in **Long-run**:

(Summary in words)

1. $\Pi > 0$:
   - Firms enter to capture some of the market profit
   - Entrant of firms increases the number of **close substitutes** in the market
   - Greater number of close substitutes will cause the **residual demand curve** to **decrease** and become **more elastic**
   - Firms continue to enter until $\Pi = 0$, $P = ATC$
2. $\Pi < 0$:
   - Firms exit the market to pursue more profitable endeavors
   - The number of **close substitutes** remaining firms face will decrease
   - **Residual demand curves** shift-**up** and become relatively more **inelastic**
   - Firms continue to exit until $\Pi = 0$, $P = ATC$

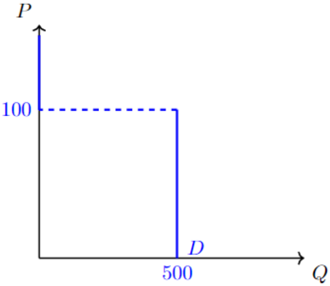
Oligopoly

- **A few** firms selling **homogeneous or differentiated** products
- Oligopolists do not face the market demand, rather a **residual demand curve**
- An oligopolist's market power arises from **barriers to entry**, but those barriers are not as prohibitive as those associated with monopoly
  - Because barriers to entry and exit exist, long-run economic profits may not necessarily be zero
- Due to a small number of firms in a market, each firm will interact with all other firms **strategically**
- We are going to proceed by looking at oligopolies producing
  - Homogeneous(identical) products
  - Differentiated products

Oligopoly

Duopoly with Identical products

Similar to Monopoly, but with consideration of the competitor's behavior



$$D = \begin{cases} 500 & \text{if } P_A < P_B \text{ and } 0 \le P_A \le 100 \\ 500/2 & \text{if } P_A = P_B \text{ and } 0 \le P_A \le 100 \\ 0 & \text{if } P_A > P_B \end{cases}$$

- We are assuming consumers want 500 units of the product as long as the price is $0 \le P \le 100$
- Firm A, B face the same residual demand
- Firms will **compete in prices**, and the low-priced producer "gets" the market (known as **Bertrand Competition**)

Oligopoly

Duopoly with Identical products

If a firm is an optimizing agent, then what is the minimum amount they would be willing to set the price at?

- Price equal to marginal cost of producing the last unit
    - If both firms have identical cost structures then $P_A = P_B = MC$ and they will split the market, $P = MR = MC$
    - If Firm A has a lower MC than they will be able to set a lower price and capture all of the market
- Therefore, each firm will compete until $P_i = MC_i$

Oligopoly

Duopoly with Differentiated products

With differentiated products,
"Which way the demand for one good affects the demand for the other and vice versa?"
– We can demonstrate this interaction between firms through a **price game**

		Firm 2: High Price	Firm 2: Low Price
Firm 1	High Price	$150,000 , $200,000	$50,000 , $300,000
	Low Price	$200,000 , $150,000	$100,000 , $175,000

- In this example, both firms have a dominant strategy in playing the "Low Price" strategy
- Therefore, there exists a dominant strategy equilibrium: (Low, Low)